

1. Synchronous protocols are always preferable to asynchronous protocols, except for the additional pin needed for clock
 - a. True
 - b. False
2. `stm32l4xx_hal_i2c.h` is a part of
 - a. CMSIS
 - b. HAL
 - c. BSP
 - d. Core
3. Which of the following is NOT a characteristic of the I2C protocol?
 - a. Serial
 - b. Synchronous
 - c. Full-duplex
 - d. In-band addressing
4. An I²C-based accelerometer gives a reading in the form of a signed 10-bit value A, in the form of two bytes with the format `xxxxxxA1A0` (LSB) and `A9A8A7A6A5A4A3A2` (MSB). The accelerometer can be configured to have ranges of ± 1 g, ± 2 g, ± 4 g, or ± 8 g, where g = acceleration due to gravity = 9.8 m/s^2 .

There is a block data update (BDU) setting in the accelerometer with the effect as described below.

0: LSB and MSB can be read independently and could belong to different samples.

1: LSB and MSB are updated only after both the MSB and LSB corresponding to a previous sample have been read.

An application requires the ability to read acceleration between -5 m/s^2 and $+5 \text{ m/s}^2$ such that a change of at least 0.1 m/s^2 is measurable. Suggest the range and BDU settings to read the acceleration in the most efficient manner, with justifications for your choice.